Power Platform & Power BI Governance

“Secure, Compliant, and Reliable Business Solutions”

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Governance Overview

Why Governance is Important?

- **Without governance:**
  - Anyone can create apps/flows without control.
  - Risk of **data leaks**, errors, and duplicate work.
  - Business processes may break unexpectedly.

- **With governance:**
  - Apps and flows are **secure and reliable**.
  - Data is **protected** and **compliant** with company policies.
  - Users can still **innovate safely**.

Governance Overview

Data Governance

- A **framework of rules, processes, and responsibilities** for managing data.

- Ensures data is:
  - **Accurate** – no duplication, errors, or outdated info.
  - **Secure** – protected from unauthorized access.
  - **Trusted** – reliable for decision-making.
  - **Compliant** – aligned with company policies and regulations.

- Helps businesses reduce risks and maintain **confidence in data quality**.

Governance Overview

Power Platform Governance

- Focused specifically on **Power Apps, Power Automate and Power BI**
- Means putting in place:
  - **Policies** – define what is allowed (e.g., connectors, sharing rules).
  - **Controls** – environments (DEV, QA, PROD) to separate work.
  - **Processes** – structured way to move apps from idea → testing → live.
- Ensures apps, flows, bots, and reports are:
  - **Built and tested safely**.
  - **Shared securely**.
  - **Managed efficiently**.
- Balances **security & compliance** with the flexibility for users to **innovate**.

Governance Framework

Roles & Responsibilities – Power Platform

Roles	Responsibilities
Admins	• Manage environments (DEV, QA, PROD) • Apply DLP (Data Loss Prevention) policies • Approve or restrict connectors (e.g., Twitter, Outlook, SharePoint)
Makers	• Create apps, flows, solutions • Develop in non-production environments (DEV/QA) • Support production deployments via pipelines under governance • Follow governance guidelines
Users	• Use apps/solutions in PROD • Accountable for business outcomes • Ensure the solution solves real business needs

Governance Framework

Roles & Responsibilities – Power BI

Roles	Responsibilities
Admins	• Manage workspace access and permissions • Add/remove users or groups • Configure workspace settings • Promote or certify datasets/reports
Members	• Edit and publish reports and dashboards • Create and manage datasets • Share content with others • Manage workspace content (except admin settings)
Contributors	• Publish new content (reports, dashboards, datasets) • Edit existing reports and dashboards • Schedule refresh and adjust datasets they own • Cannot manage permissions
Viewers	• View reports and dashboards only • Interact with filters, slicers, and visuals • Export data if allowed • Cannot edit, publish, or share content

Environment Strategy

Power Platform environment:

It is designed for business users to safely build, test, and use apps, flows, and reports.

Purpose: To provide a safe, reliable, and compliant space for building solutions without risking the company's data or live systems.

i. **Citizen-led Development**
ii. **RPA (Bot) Environment**
iii. **AI Environment**

Environment Strategy

Enterprise Environment (Citizen-led Development for standard connectors)

- **Purpose:** Provides a **safe, governed workspace** for business users (citizen developers) to create solutions.

- **Who uses it?**
  - Business users / Makers.

- **Structure:**

Type	Environment Name	Region	Purpose
DEV	PP-DEV-US	US	Space to experiment and build apps/flows (prototyping, innovation)
QA	PP-QA-US	US	Validate apps with testers & business users before PROD
PROD	PP-PROD-US	US	Stable, live environment with only approved apps/flows

Environment Naming Standard

Component	Description
PP	Power Platform
DEV/QA/PROD	Environment Type
US	Region

Note: *If a new region is required, a formal request needs to be submitted before creation.*

Environment Strategy

Enterprise Environment (Citizen-led Development for premium connectors)

- **Purpose:** Provides a **safe, governed workspace** for business users (citizen developers) to create solutions using premium connectors.

What are Premium Connectors? *Special connections that let apps securely link to enterprise systems or external services via APIs (e.g., SQL, Salesforce, SAP, HTTP).*

- **Who uses it?**
  - Business users / Makers.

- **Structure:**

Type	Environment Name	Region	Purpose
DEV	PP-DEV-US (PRM)	US	Space to experiment and build apps/flows (prototyping, innovation)
QA	PP-QA-US (PRM)	US	Validate apps with testers & business users before PROD
PROD	PP-PROD-US (PRM)	US	Stable, live environment with only approved apps/flows

Environment Naming Standard

Component	Description
PP	Power Platform
DEV/QA/PROD	Environment Type
US	Region
PRM	Premium

Note: *If a new region is required, a formal request needs to be submitted before creation.*

Environment Strategy

Enterprise Environment - RPA (Bot) Environment

- **Purpose:** Provides a **dedicated space for Robotic Process Automation (RPA)** solutions.

- **Who uses it?**
  - Developers who design unattended bots (bots that run on their own).

- **Structure:**

Type	Environment Name	Region	Purpose
DEV	PP-RPA-DEV-US	US	Build and test automation bots with test data
QA	PP-RPA-QA-US	US	Test bots for reliability & simulate real-world conditions
PROD	PP-RPA-PROD-US	US	Fully tested bots run business processes in production

Environment Naming Standard

Component	Description
PP	Power Platform
RPA	Solution / Purpose
DEV/QA/PROD	Environment Type
US	Region

Note: *If a new region is required, a formal request needs to be submitted before creation.*

Environment Strategy

Enterprise Environment - AI Environment

- **Purpose:** Dedicated space for **AI and machine learning solutions** within Power Platform.

- **Who uses it?**
  - AI developers, data scientists, or makers experimenting with AI models.

- **Structure:**

Type	Environment Name	Region	Purpose
DEV	PP-AI-DEV-APAC	ASIA	Experiment with AI models; safe space for makers to try new AI-based apps
QA	PP-AI-QA-APAC	ASIA	Test AI models using sample/test data; validate accuracy, performance, reliability
PROD	PP-AI-PROD-APAC	ASIA	Stable environment for approved AI-driven apps in production

Environment Naming Standard

Component	Description
PP	Power Platform
AI	Solution / Purpose
DEV/QA/PROD	Environment Type
APAC	Region

Note: *If a new region is required, a formal request needs to be submitted before creation.*

DLP (Data Loss Prevention) Policies

What is DLP?

- **Data Loss Prevention (DLP)** ensures sensitive data does not move across untrusted or unauthorized services.
- It enforces rules on which **connectors** can be used together.

Where is it applied?

- Applied at the **environment level**
- Defines **business data vs. non-business data** connectors
- Prevents data leakage between services (e.g., Outlook → Twitter)
- Helps enforce compliance and protect organizational data

DLP (Data Loss Prevention) Policies

Why DLP is Needed?

- **Protects Sensitive Data** – Prevents confidential information from leaking.
- **Ensures Compliance** – Meets industry regulations (GDPR, HIPAA, PCI-DSS).
- **Reduces Insider Risks** – Stops accidental or malicious data sharing.
- **Avoids Financial & Reputation Loss** – Prevents fines and loss of client trust.
- **Secures Remote & Cloud Work** – Safeguards data across devices and cloud apps.
- **Monitors & Controls Data Use** – Provides visibility and control over data flow.

DLP (Data Loss Prevention) Policies

Data Loss Prevention (DLP) Connectors

Connectors classified into **Business** vs. **Non-Business**:

- **Business Data Only:** SharePoint, Dataverse, Outlook, etc.
- **Non-Business Data:** Twitter, Gmail, social media, etc.

Environment Type	Allowed Connectors	Connector Category
Citizen-Led	All Standard Connectors	E.g., SharePoint, Outlook, Teams, OneDrive
RPA (Robotic Process Automation)	Desktop / Premium Connectors	E.g., UI Flows, Excel Desktop
AI (Artificial Intelligence)	Premium/ Built-in Copilot Studio Connectors	E.g., AI Builder, Dataverse, Copilot Studio

Notes / Guidelines:

1.Applied at environment level *– ensures connectors comply with organizational policies per environment.*
2.Prevents data leakage *– stops business data from moving to untrusted services. E.g.: You cannot send HR data to Gmail or Twitter.*
3.Intake Process *– any new connector must be requested and justified via IT.*

Security & Access Control

How Access is Managed

- **Access Packages** → Users request access via packages, ensuring approvals are tracked and standardized.
- **Least Privilege Access** → Users only get the minimum rights needed to perform their role.
- **Azure AD Groups (RBAC)** → Roles and permissions assigned via AD groups for consistency and easier management.

Security & Access Control

Role-Based Security Model

Platform	Role	Scope / Environment	Access Level
Power Platform	Admins	All environments (DEV, QA, PROD)	Full control: manage environments, security, policies, connectors
	Makers	DEV, QA, PROD	Build & test in DEV/QA; restricted roles in PROD (deployment purpose, limited edits)
	Users	DEV, QA, PROD (via groups)	DEV: Environment access QA: Validation/testing PROD: Use approved apps/flows
Power BI	Admins	All workspaces	Full control: manage permissions, settings, and content across all workspaces
	Members	Dedicated per workspace	Create, edit, publish, and manage content within assigned workspace
	Contributors	Dedicated per workspace	Publish/edit reports & datasets; cannot manage permissions
	Viewers	Dedicated per workspace	View & interact with reports/dashboards only

Security & Access Control

Citizen-led Environment AD Groups

The following Azure AD groups control access for citizen developers and users

Group Name	Role	Environment Scope
AZ-PP-ALL-ADMN	Admin	DEV, QA & PROD
AZ-PP-US-MKR-NONPROD	Maker	DEV & QA
AZ-PP-US-USR-NONPROD	App User	DEV & QA
AZ-PP-US-MKR	Maker	PROD

Security & Access Control

Citizen-led premium Environment AD Groups

The following Azure AD groups control access for citizen developers and users

Group Name	Role	Environment Scope
AZ-PP-ALL-ADMN	Admin	DEV, QA & PROD
AZ-PP-US-MKR-NONPROD-PRM	Maker	DEV & QA
AZ-PP-US-USR-NONPROD-PRM	App User	DEV & QA
AZ-PP-US-MKR-PRM	Maker	PROD

Security & Access Control

RPA Environment AD Groups

Group Name	Role	Environment Scope
AZ-PP-RPA-ADMN	Admin	DEV, QA & PROD
AZ-PP-US-RPA-USR	App User	PROD
AZ-PP-US-RPA-MKR-NONPROD	Maker	Dev & QA
AZ-PP-US-RPA-USR-NONPROD	App User	Dev & QA

Note: Admins, Makers, and Users must all be part of the environment's user group..

Security & Access Control

AI Environment AD Groups

Group Name	Role	Environment Scope
AZ-PP-AI-ADMIN	AI Admin	DEV, QA & PROD
AZ-PP-APAC-AI-USR	AI App User	PROD
AZ-PP-APAC-AI-MKR	AI Maker	PROD
AZ-PP-APAC-AI-MKR-NONPROD	AI Maker	Dev & QA
AZ-PP-APAC-AI-USR-NONPROD	AI App User	Dev & QA

Note: Admins, Makers, and Users must all be part of the environment's user group..

Security & Access Control

PowerBI Workspace AD Groups

Workspace Name	Admin Group	Member Group	Viewer Group	Contributor Group
WorkspaceExample_GLBL	AZ-PBI-ALL-ADMN	AZ-Group-MBR	AZ-Group-VIEWER	AZ-Group-CONTRIB

Application Lifecycle Management (ALM)

What is ALM?

ALM is the process of **managing an app or report from development to production**, including planning, building, testing, deploying, and monitoring.

Why is it important?

- Ensures quality, compliance, and controlled deployments.
- Reduces risks of errors or downtime.
- Supports traceability and governance across environments.

Process

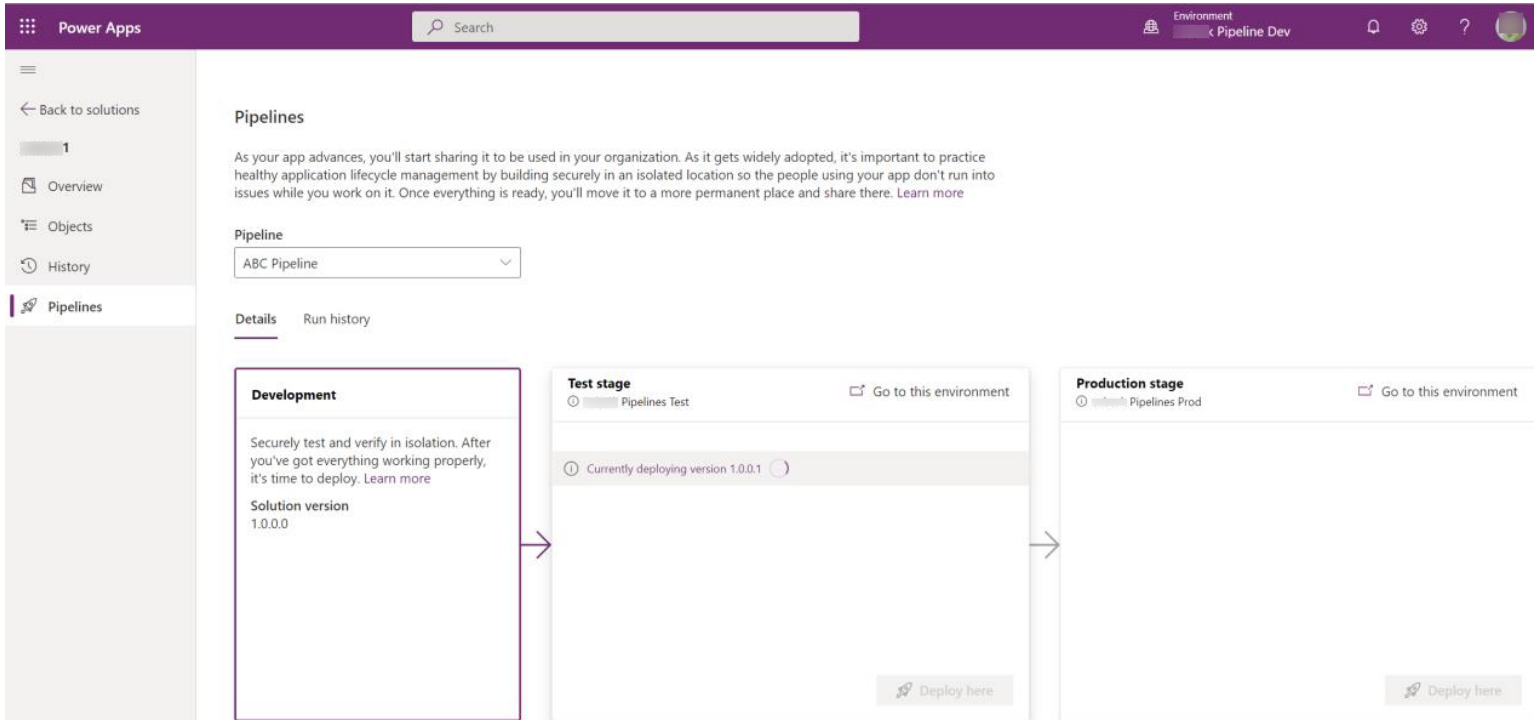
- **Plan & Build** – DEV environment; makers create apps/reports.
- **Deploy & Release** – PROD environment; approved solutions go live.
- **Monitor & Maintain** – Track usage, performance, and issues.
- **Test & Validate** – QA environment; testing, bug fixes, validation.

Note: Applies to both Power Platform apps/flows/bots and Power BI reports/datasets.

Power Platform (ALM)

ALM ensures solutions are built, tested, deployed, and monitored across DEV → QA → PROD in a controlled and compliant manner.

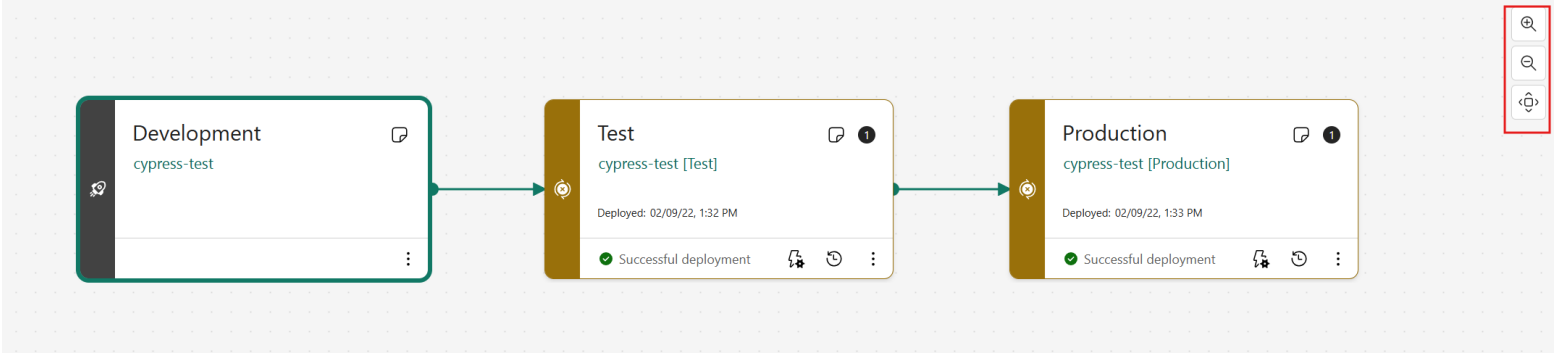
- **DEV:** Makers build apps, flows, and bots.
- **QA:** Testers and business users validate functionality, performance, and compliance.
- **PROD:** Deploy approved solutions via controlled pipelines.
- **Monitor:** Track usage, detect issues, and apply updates.
- **Governance:** All stages follow policies; pipelines ensure secure, consistent deployments.



PowerBI (ALM)

ALM ensures solutions are built, tested, deployed, and monitored across DEV → TEST → PROD in a controlled and compliant manner.

- **DEV Workspace:** Create reports and datasets.
- **QA Workspace:** Validate visuals, data refreshes, and accuracy.
- **PROD Workspace:** Deploy approved reports/datasets using pipelines.
- **Monitor:** Track report usage and data refresh status.
- **Governance:** Pipelines and workspace policies ensure controlled promotion and compliance



Data Sources & Power BI SharePoint

Data Sources

- Power BI reports and dashboards are built from **multiple data sources** (databases, Excel files, cloud apps, SQL servers, etc.).
- These sources feed data into **Power BI Workspaces** for reporting and analysis.

Power BI SharePoint Integration

- Each **Power BI Workspace** has its own **dedicated SharePoint site**.
- Access is **role-based**:
  - ✓ **Admins** → Full control (manage reports, permissions, settings).
  - ✓ **Members** → Edit content, read/write data.
  - ✓ **Contributors** → Contribute to access data for building reports.
- This structure ensures **secure, organized, and role-specific collaboration**.

Data Sources & Power BI SharePoint

Key Benefits

- **Role-based secured access** → Right people get the right access.
- **Organized storage per workspace** → Each workspace data stays separate
- **Team collaboration** → Members can work together safely.
- **Seamless integration** → Smooth link between **Power BI** and **SharePoint**.

Center of Excellence (COE)

What is COE?

- A dedicated hub/team for governance, standards, and innovation
- Ensures best practices and alignment across business & IT
- Acts as a central body for monitoring, policies, and enablement

Why Do We Need a COE?

- Standardize processes & enforce governance
- Ensure compliance and security across platforms
- Provide training, support & reusable resources
- Reduce risks and accelerate innovation

Center of Excellence (COE)

How COE Helps in Monitoring

- Tracks adoption, usage & performance of apps/reports
- Monitors compliance with policies & DLP rules
- Provides dashboards for leadership visibility
- Identifies unused or duplicate assets to optimize cost

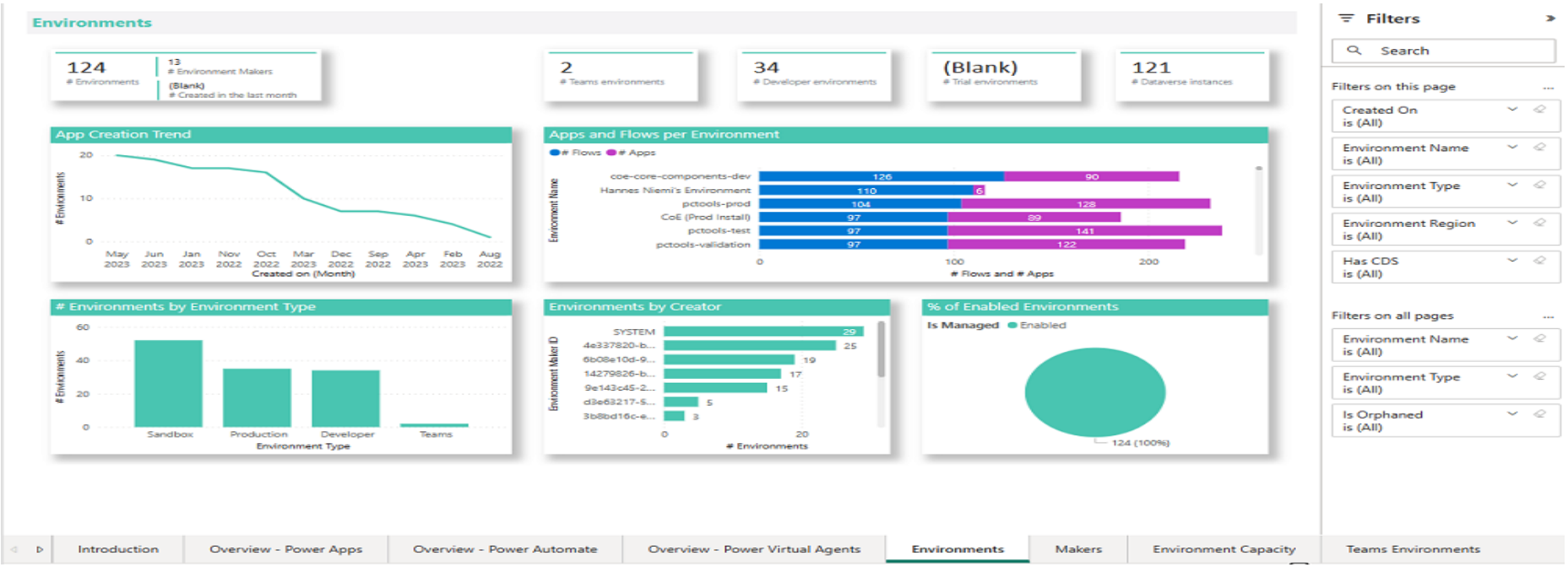
How you should monitor

- Governance dashboards (apps, flows, environments, licenses)
- Role-based access monitoring (Admin/Member/Viewer)
- Policy enforcement for security & compliance
- Lifecycle management: creation → updates → retirement

Monitoring & Auditing

- Use **CoE (Center of Excellence) Toolkit** to:
  - Monitor all apps, flows, bots across environments.
  - Track compliance and security risks.
  - Audit usage and changes for accountability.

- **COE Dashboard**



Thank You!